
Problem

```
x_old = [-1;-2;1];
D_inv = jac_i(x_old);
Fx = F(x_old);
x = x_old - D_inv*Fx; %Update as per Newton's Method for first time
k=1; %Set iteration counter
Fx = F(x); %Evaluate function at new variable
fprintf("k\t x1\t\t x2\t\t x3\t\t ||dx||_inf\t ||F(x)||_inf\n");
fprintf("%d\t %.6f\t %.6f\t %.6f\t %.2e\t %.2e\n",...
        k,x(1),x(2),x(3),norm(x-x_old,"inf"),norm(Fx,"inf"));
while(norm(x-x_old,"inf")>=1e-8) %Keep iterating while error is more than 1e-8
    x_old = x;
    D_inv = jac_i(x_old);
    Fx = F(x_old);
    x = x_old - D_inv*Fx;
    Fx = F(x);
    k = k+1;
    fprintf("%d\t %.6f\t %.6f\t %.6f\t %.2e\t %.2e\n",...
            k,x(1),x(2),x(3),norm(x-x_old,"inf"),norm(Fx,"inf"));
end
fprintf("x = [%.6f %.6f %.6f]^T\n",x(1),x(2),x(3));
%fprintf("%.16f %.16f %.16f",Fx(1),Fx(2),Fx(3));

function D_inv = jac_i(x) %Creates Jacobian Matrix evaluated at a point
    D(:,1)= [3*x(1)^2+2*x(1)*x(2)-x(3); exp(x(1)); -2*x(3)];
    D(:,2)= [x(1)^2;exp(x(2));2*x(2)];
    D(:,3)= [-x(1); -1; -2*x(1)];
    D_inv = inv(D);
end

function y = F(x) %Evaluates given function at some arbitrary vector say x
    y = [x(1)^3+(x(1)^2)*x(2)-x(1)*x(3)+6;exp(x(1))+exp(x(2))-
x(3);x(2)^2-2*x(1)*x(3)-4];
end

k  x1    x2    x3    ||dx||_inf  ||F(x)||_inf
1  -1.636738 -1.514277 0.334707 6.65e-01 1.89e+00
2  -1.464857 -1.668232 0.414166 1.72e-01 1.16e-01
3  -1.456107 -1.664191 0.422475 8.75e-03 6.39e-04
4  -1.456043 -1.664230 0.422493 6.46e-05 1.89e-08
5  -1.456043 -1.664230 0.422493 1.56e-09 1.78e-15
x = [-1.456043 -1.664230 0.422493]^T
```

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